AI/ML CAPSTONE / INDIVIDUAL PROJECT

TASHKENT APARTMENT PRICE PREDICTOR

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PROJECT IN ONE LINE

Estimate a current August 2026 Tashkent apartment asking price from observable listing attributes.

Reference estimate — not an appraisal

4,214	RF	0.681	$27,195
MODELING ROWS	SELECTED MODEL	TEST R^2	TEST MAE

WHO NEEDS IT — AND WHAT DOES ML DO?

Lock the user, input, output, and boundary in under 45 seconds.

USER

Buyers, sellers, agents, and analysts who need a consistent August 2026 reference estimate.

SUPERVISED REGRESSION

Inputs: district, size, rooms, apartment level, building levels, new-build/resale.

Target: asking price.

OUTPUT + BOUNDARY

One estimated August 2026 asking price in USD, plus warnings.

Not a completed sale price, guarantee, or legal appraisal.

RAW APARTMENT → VALIDATED PIPELINE → USD ESTIMATE + WARNING

FROM 4,867 LISTINGS TO A GROUP-SAFE TEST

Privacy-minimized public HATA snapshot from August 2026.

SOURCE

4,867 parsed
11 districts

→

CLEAN

257 invalid + 396 duplicates removed

→

SPLIT

3,840 groups
80% / 20%

→

SELECT

5-fold CV
group-safe

→

EVALUATE

835 unseen rows
768 groups

Leakage controls

feature+target deduplication • identical fingerprints grouped • no price/m^2 leakage • preprocessing inside CV • test not used for selection

FOUR APPROACHES — ONE EVIDENCE-BASED CHOICE

Final model selected by the lowest five-fold development CV MAE, not test performance.

MODEL	CV MAE	CV RMSE	CV R^2
Median baseline	$55,604	$123,587	-0.068
Log Ridge	$38,301	$105,755	0.219
Random Forest	**$31,298**	**$80,394**	**0.556**
Gradient Boosting	$33,545	$94,081	0.396

WHY RANDOM FOREST?

Best validation MAE

Captures non-linear size × location interactions

Trade-off: larger and less interpretable than Ridge

UNSEEN-TEST RESULT — WITH THE FAILURE VISIBLE

The baseline comparison and error case matter more than a polished success claim.

$27,195	$58,887	0.681	24.58%	−46.6%
MAE	RMSE	R^2	MAPE	MAE vs baseline

LARGEST ERROR

Shayhontohur • 160 m^2 • 3 rooms • new build
Actual: $1,000,000 Predicted: $234,461
Absolute error: $765,539

WHAT IT TEACHES

Luxury premiums and condition are not observed. Mirobod and Shayhontohur MAE is high; some district slice R^2 values are negative.

ONE REAL INPUT → ONE VISIBLE RESULT

Colab route: setup → load saved pipeline → predict → show validation error.

RAW INPUT

District	Chilonzor
Size	70 m^2
Rooms	3
Level	3 / 5
Building type	Resale

→

ESTIMATED AUGUST 2026 ASKING PRICE

$97,098 USD

EDGE CASE

Level 9 / max 5 → clear ValueError

Open: README Colab badge • Run all • Do not tour source code unless asked

HONEST LIMITS — CLEAR NEXT STEP

A useful educational model is stronger when its boundary is explicit.

LIMITATIONS

Dated asking-price snapshot
No condition / exact building / legal status
Unequal district reliability
Source noise and market drift

SAFE USE

August 2026 reference only
Compare with recent listings
Require human review
No lending, tax, or legal appraisal

NEXT IMPROVEMENT

Collect verified transactions with exact neighborhood, condition, building year, renovation, and legal status; then use a later time holdout.

Thank you. I am ready for one evidence-based question.

OFFICIAL CRITERIA → EXACT REPOSITORY PROOF

Open docs/capstone_evidence_matrix.md for the complete claim → location → why-proof mapping.

1 Problem

README + Project Brief

2 Data

data/README + EDA + src/data.py

3 Models

model_comparison.csv + src/model.py

4 Evaluation

test metrics + errors + district slices

5 Delivery

demo.ipynb + saved joblib + predict.py

6 Reproduce

README + clean_run_check + CI

7 Responsible

README limitations + data card

8 Defense

deck + pitch + question bank

YELLOW only where external approval/rehearsal evidence is still required.

THE SAME PIPELINE TRAINS AND PREDICTS

Preprocessing is serialized with the estimator, preventing notebook-only inference drift.

RAW INPUT

district • rooms • size • level • max_levels • new-build/resale

VALIDATE

required fields • positive values • level ≤ max_levels

FEATURE

floor_ratio = level / max_levels

PREPROCESS

OneHotEncoder district • numeric passthrough / scaling

MODEL

selected Random Forest inside sklearn Pipeline

OUTPUT

USD estimate + unseen/range warnings

Source proof: src/data.py + src/model.py + artifacts/house_price_pipeline.joblib

ANSWER WITHOUT GUESSING

Direct answer → exact evidence → limitation or next step.

WHY RANDOM FOREST?

Lowest group-safe CV MAE ($31,298).
Open reports/model_comparison.csv.
Trade-off: size and interpretability.

IS IT CURRENT?

Yes — dated Aug 2026 asking-price snapshot.
Open data/README.md.
Not a permanently live or completed-sale model.

BIGGEST FAILURE?

$1m Shayhontohur actual vs ~$234k predicted.
Open largest_errors.csv.
Missing luxury/condition signal.

Full evidence-anchored answers: docs/defense_question_bank.md